

Total Marks of
Question no.

Examiner

Moderator

Re-Assessor

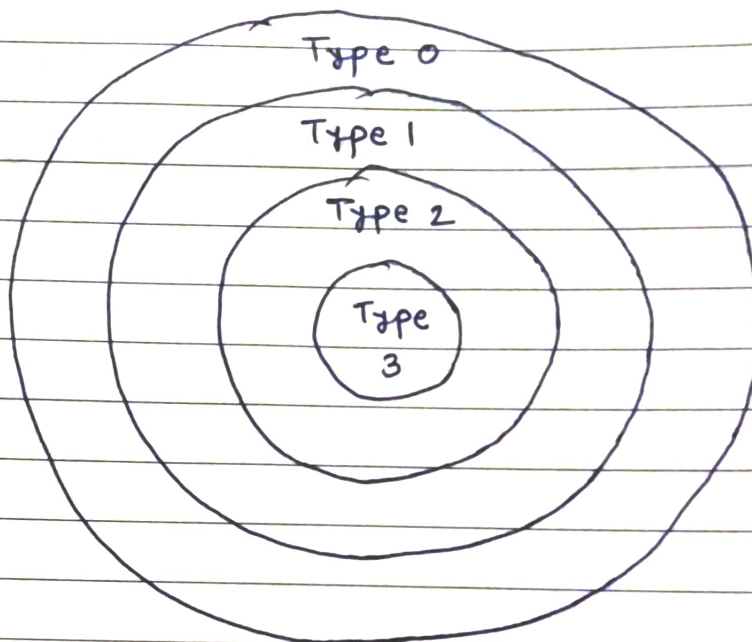
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Q. 1

A) Types of grammar :-



Type 0 : Unrestricted grammar

Language recognized by TM is known as Type 0 grammar.

It is also known as recursively enumerable lang.

Production is given by

$$\alpha \rightarrow \beta$$

where α & β are in sentential form
e.g.

$$Sba \rightarrow a$$

$$S \rightarrow B$$

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Type-1 : Context Sensitive grammar

This grammar is recognized by linear bound automata.

This grammar represents context sensitive lang.

Prodⁿ is in the form of $\alpha \rightarrow \beta$

Where LHS must be less than or equal to RHS.

e.g.

$$S \rightarrow BA$$
$$BA \rightarrow bca$$
$$B \rightarrow b$$

Type-2 : Context Free grammar

This grammar is recognized by push down automata.

This grammar represents context free lang.

Prodⁿ is in the form of $A \rightarrow \alpha$
A is variable & α is in sentential form.

e.g. $A \rightarrow aA \mid b \mid Bb$

$$B \rightarrow a \mid b$$

Type-3 : Regular grammar

This grammar is recognised by finite automata

This grammar represents regular lang.

Prodⁿ is in the form of $V \rightarrow T^* V / T^*$
e.g.

$$A \rightarrow ab$$

Q.1 B) Moore m/c	Mealy m/c
1) It is a finite automata with o/p but no final state	It is a finite automata with o/p but no final state
2) state gives o/p	state & i/p gives o/p
3) In transit ⁿ dgm, o/p associated with state	In transit ⁿ dgm, o/p associated with i/p
4) o/p mapping f ⁿ is: $\lambda: Q \rightarrow \Delta$	o/p mapping f ⁿ is: $\lambda: Q \times Z \rightarrow \Delta$
5) If i/p length is n then o/p length is $n+1$	i/p & o/p length is n

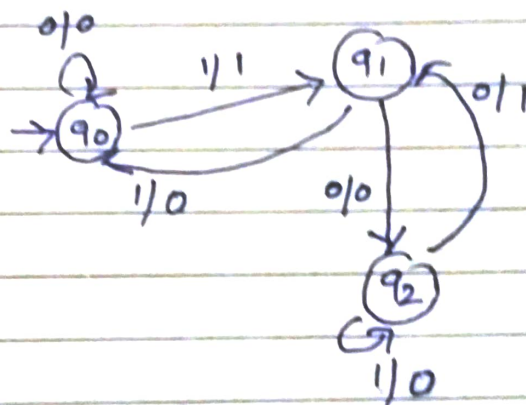
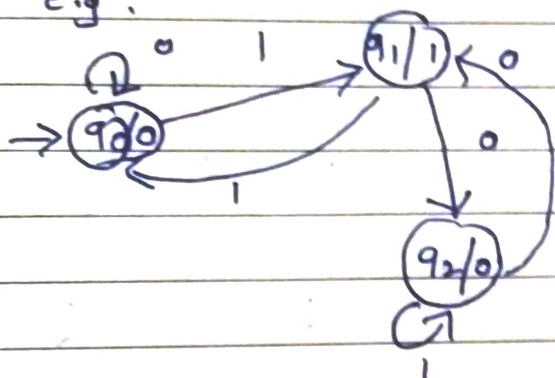
6) It may have more states than Mealy m/c

It may be less states than Moore m/c

7) More logic is required to decode the o/p resulting in more circuit delays.

It react faster to i/p's, generally in the same clock cycle

8) e.g.



Q.1

c) 1) LMD

$S \rightarrow aB$

$\rightarrow aaBB$

$\rightarrow aabSB$

$\rightarrow aabaBB$

$\rightarrow aababbB$

$\rightarrow aababbbs$

$\rightarrow aababbbaB$

$\rightarrow aababbbaab$

2) RMD

$S \rightarrow aB$

$\rightarrow aaBB$

$\rightarrow aabbs$

$\rightarrow aabbaB$

$\rightarrow aabbaab$

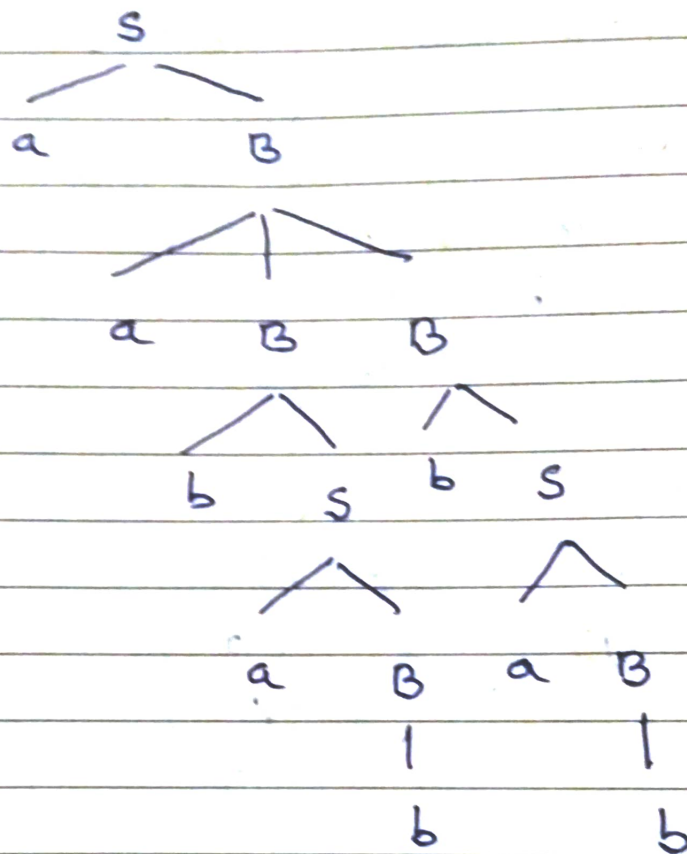
$\rightarrow aabsbab$

$\rightarrow aabaBbaab$

$\rightarrow aababbbaab$

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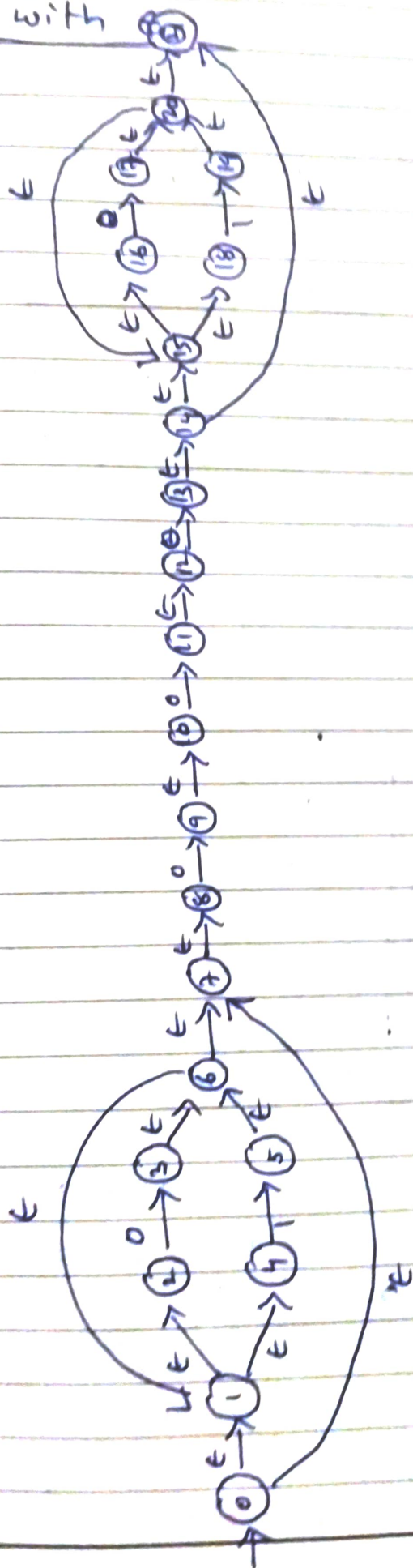
Parse Tree



Q.2 Regular expⁿ is
 $(0+1)^* 000 (0+1)^*$

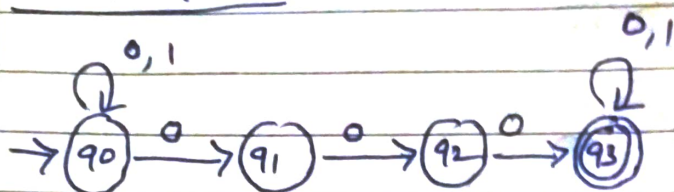
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NFA with



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NFA w/o ϵ NFA to DFA conversionLet $M = (Q, \Sigma, \delta, q_0, F)$ be the DFA $Q = \{ [q_0], [q_0, q_1], [q_0, q_1, q_2], [q_0, q_1, q_2, q_3] \}$ $\Sigma = \{ 0, 1 \}$ $q_0 : [q_0]$ $F = \{ [q_0, q_1, q_2, q_3] \}$ $\delta :$

$Q \backslash \Sigma$	0	1
$[q_0]$	$[q_0, q_1]$	$[q_0]$
$[q_0, q_1]$	$[q_0, q_1, q_2]$	$[q_0]$
$[q_0, q_1, q_2]$	$[q_0, q_1, q_2, q_3]$	$[q_0]$
* $[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_3]$
* $[q_0, q_3]$	$[q_0, q_1, q_3]$	$[q_0, q_3]$
* $[q_0, q_1, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_3]$

Merge $[q_0, q_1, q_2, q_3]$ & $[q_0, q_1, q_3]$ rename $\rightarrow [q_0, q_1, q_2, q_3]$

$Q \backslash \Sigma$	0	1
$[q_0]$	$[q_0, q_1]$	$[q_0]$
$[q_0, q_1]$	$[q_0, q_1, q_2]$	$[q_0]$
$[q_0, q_1, q_2]$	$[q_0, q_1, q_2, q_3]$	$[q_0]$
* $[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_3]$
* $[q_0, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_3]$

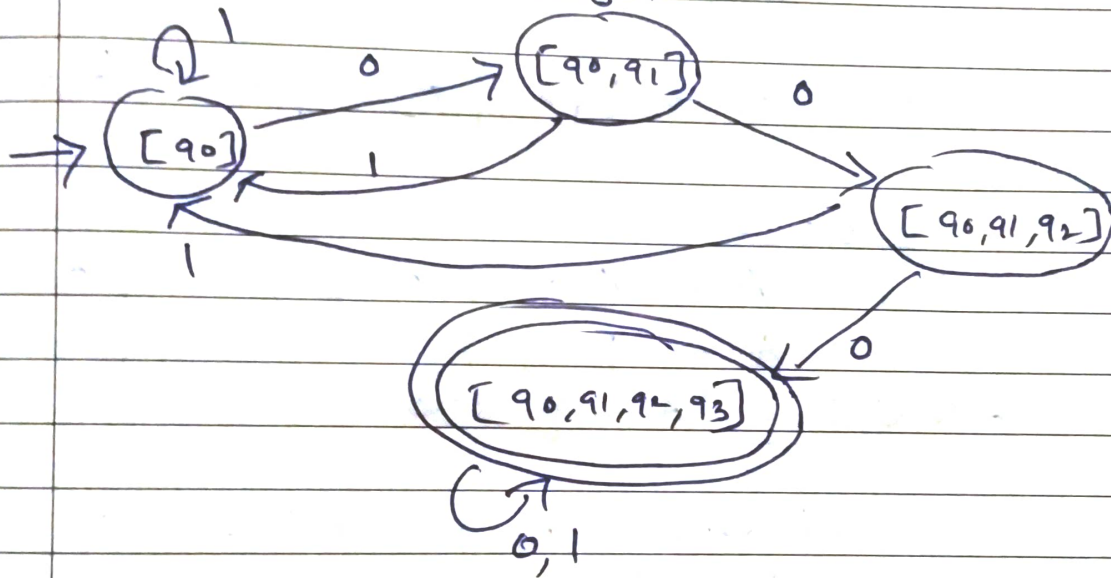
Merge

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Σ	0	1
$[q_0]$	$[q_0, q_1]$	$[q_0]$
$[q_0, q_1]$	$[q_0, q_1, q_2]$	$[q_0]$
$[q_0, q_1, q_2]$	$[q_0, q_1, q_2, q_3]$	$[q_0]$
* $[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$	$[q_0, q_1, q_2, q_3]$

state transitⁿ dgm

Q.2

B)

Any ϵ -free CFL can be converted into CFG that has prodⁿ in the form of

$$A \rightarrow a\alpha$$

where a be the terminal

α be the string of variables
(can be empty).

Steps :-

1) simplify the CFG

a) Removal of useless symbols

b) ——— ϵ -prodⁿ

c) ——— unit prodⁿ

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2) Apply rule 1 & rule 2 to convert
CFG into GNF.

Rule 1:

Let $A \rightarrow B\alpha$ &

$B \rightarrow B_1 \mid B_2 \mid \dots \mid B_n$

then $A \rightarrow B_1\alpha \mid B_2\alpha \mid \dots \mid B_n\alpha$

Rule 2:

Let $A \rightarrow A\alpha_1 \mid A\alpha_2 \mid \dots \mid A\alpha_k$ &

$A \rightarrow B_1 \mid B_2 \mid \dots \mid B_s$ be the
remaining A-prods

Introduce new variable B, such that

$B \rightarrow \alpha_i \mid \alpha_i B$ &

$A \rightarrow B_1 \mid B_1 B$

Given pbm

$A_1 \rightarrow A_2 A_3$

$A_2 \rightarrow A_3 A_1 \mid 0$

$A_3 \rightarrow A_1 A_2 \mid 1$

$\Rightarrow$

Step 1:-

Given grammar already in
simplified form.

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Step 2 :-

$$\therefore A_2 \rightarrow A_1 A_2 \mid 1$$

$$\therefore A_3 \rightarrow A_2 A_3 A_2 \mid 1$$

$$\therefore \underbrace{A_3}_{\alpha} \rightarrow \underbrace{A_3 A_1 A_3 A_2}_{\beta_1} \mid \underbrace{0 A_3 A_2}_{\beta_2} \mid 1$$

Introduce new variable B such that
 $B \rightarrow \alpha \mid \alpha B$

$$\therefore B \rightarrow A_1 A_3 A_2 \mid A_1 A_3 A_2 B \quad \&$$

$$A_3 \rightarrow \beta_1 \mid \beta_2 B$$

$$\therefore A_3 \rightarrow 0 A_3 A_2 \mid 1 \mid 0 A_3 A_2 B \mid 1 B \text{ in GNF. } \textcircled{1}$$

$$\therefore A_2 \rightarrow A_3 A_1 \mid 0$$

$$\therefore A_2 \rightarrow 0 A_3 A_2 A_1 \mid 1 A_1 \mid 0 A_3 A_2 B A_1 \mid 1 B A_1 \mid 0 \text{ in GNF. } \textcircled{2}$$

$$\therefore A_1 \rightarrow A_2 A_3$$

$$\therefore A_1 \rightarrow 0 A_3 A_2 A_1 A_3 \mid 1 A_1 A_3 \mid 0 A_3 A_2 B A_1 A_3 \mid 1 B A_1 A_3 \mid 0 A_3 \text{ in GNF. } \textcircled{3}$$

$$\therefore B \rightarrow A_1 A_3 A_2 \mid A_1 A_3 A_2 B$$

$$\therefore B \rightarrow 0 A_3 A_2 A_1 A_3 A_3 A_2 \mid 1 A_1 A_3 A_3 A_2 \mid 0 A_3 A_2 B A_1 A_3 A_3 A_2 \mid 1 B A_1 A_3 A_3 A_2 \mid 0 A_3 A_3 A_2 \mid 0 A_3 A_2 A A_3 A_3 A_2 B \mid 1 A_1 A_3 A_3 A_2 B \mid 0 A_3 A_2 B A_1 A_3 A_3 A_2 B \mid 1 B A_1 A_3 A_3 A_2 B \mid 0 A_3 A_3 A_2 B \text{ in GNF. } \textcircled{4}$$

Let $G = (V, T, P, S)$ be the CFG in GNF

$$V = \{ A_1, A_2, A_3, B \}$$

$$T = \{ 0, 1 \}$$

$$P = \text{prodns } (1), (2), (3) \text{ \& } (4)$$

$$S: A_1$$

Q.3A) Let $P = (Q, \Sigma, \Gamma, z_0, \delta, q_0, F)$ be the PDA

$$Q = \{$$

$$\Sigma = \{ a, b \}$$

$$\Gamma = \{ a, b, z_0 \}$$

z_0 : initial stack symbol

q_0 : initial state

$$F = q_f$$

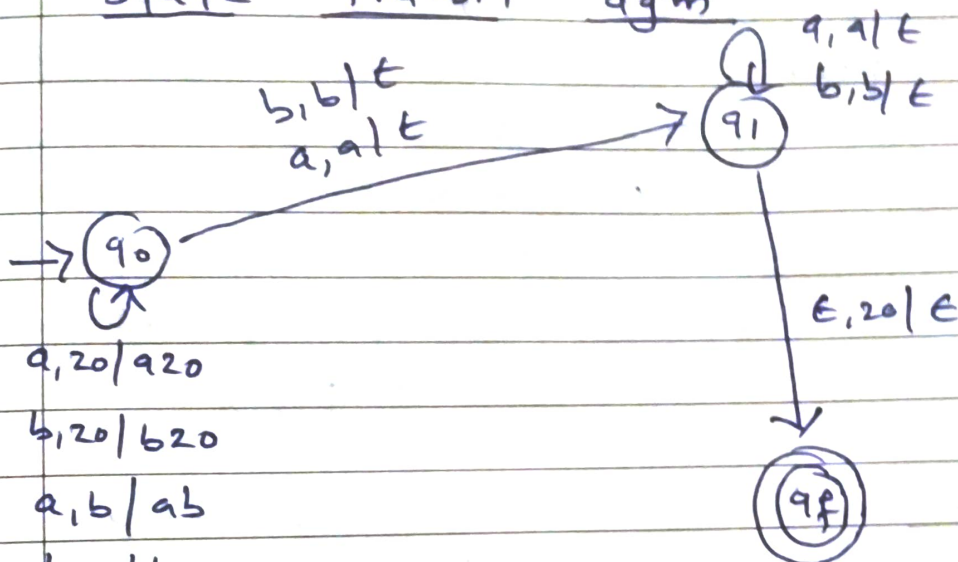
δ :

δ :	stack	Action
$\delta(q_0, a, z_0)$	(q_0, az_0)	push a
$\delta(q_0, b, z_0)$	(q_0, bz_0)	push b
$\delta(q_0, a, b)$	(q_0, ab)	push a
$\delta(q_0, b, a)$	(q_0, ba)	push b
$\delta(q_0, a, a)$	$\{ (q_0, aa), (q_1, \epsilon) \}$	push a or pop a
$\delta(q_0, b, b)$	$\{ (q_0, bb), (q_1, \epsilon) \}$	push b or pop b
$\delta(q_1, a, a)$	(q_1, ϵ)	pop a
$\delta(q_1, b, b)$	(q_1, ϵ)	pop b
$\delta(q_1, \epsilon, z_0)$	(q_f, ϵ)	pop z_0 , Accept

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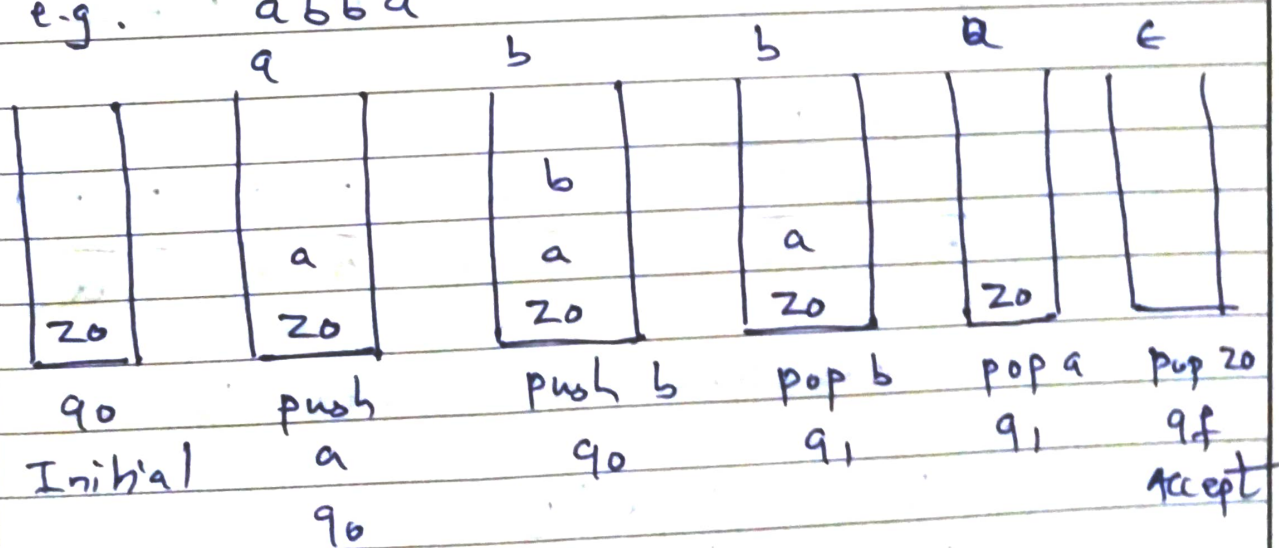
All other transitions result in error.

state transition dgm



$a, z_0 / a z_0$
 $b, z_0 / b z_0$
 $a, b / a b$
 $b, a / b a$
 $a, a / a a$
 $b, b / b b$

e.g. $a b b a$



B) CFG to PDA

Step 1: Convert CFG into GNF.

prodⁿ said to be in GNF if it is in the form of

$$A \rightarrow a\alpha$$

where a be the terminal &

α be the string of variables

(can be empty)

Step 2: Rules to convert CFG into PDA

1) $S(q_0, \epsilon, z_0) \rightarrow (q_1, Sz_0)$ push start var. S

2) $S(q_1, a, A) \rightarrow \{ (q_1, \alpha) \mid A \rightarrow a\alpha \}$

3) $S(q_1, \epsilon, z_0) \rightarrow (q_f, \epsilon)$ pop z_0 , Accept

Given pbm

$$\begin{array}{c} E \rightarrow E + E \mid E - E \mid E * E \mid E / E \mid id \\ \hline \overline{A} \quad \overline{A} \quad \overline{\alpha_1} \quad \overline{A} \quad \overline{\alpha_2} \quad \overline{A} \quad \overline{\alpha_3} \quad \overline{A} \quad \overline{\alpha_4} \quad \overline{B} \end{array}$$

Introduce new var B such that

$$B \rightarrow \alpha_i \mid \alpha_i B$$

$$\therefore B \rightarrow +E \mid -E \mid *E \mid /E \mid +EB \mid -EB \mid$$

$$*EB \mid /EB \text{ in GNF}$$

$$E \rightarrow B \mid B B$$

$$\therefore E \rightarrow id \mid id B \text{ in GNF.}$$

Let $P = (Q, \Sigma, \Gamma, z_0, \delta, q_0, F)$ be the PDA

$Q = \{q_0, q_1, q_f\}$

$\Sigma = \{+, -, *, /, id\}$

$\Gamma =$

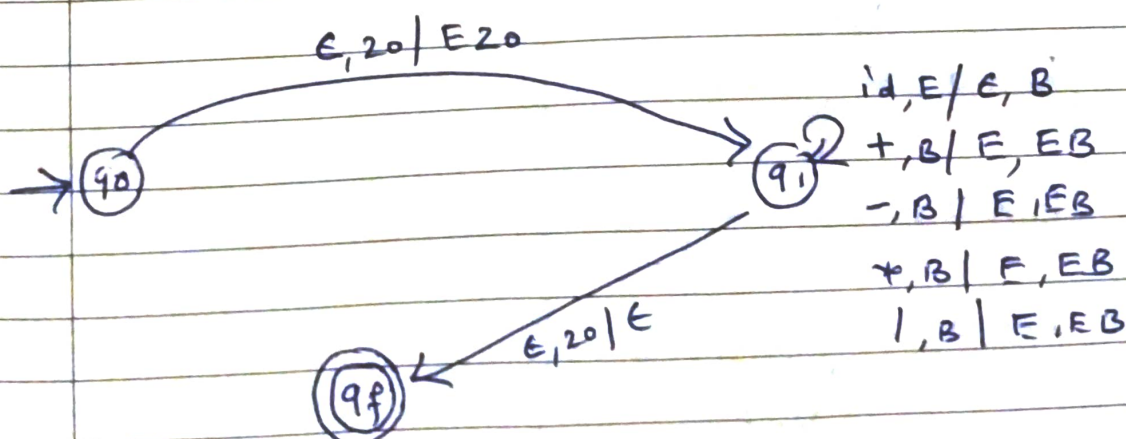
z_0 : initial stack symbol

q_0 : initial state

$F = \{q_f\}$

δ	stack
$\delta(q_0, \epsilon, z_0)$	$(q_1, \epsilon z_0)$
$\delta(q_1, id, \epsilon)$	$\{(q_1, \epsilon), (q_1, B)\}$
$\delta(q_1, +, B)$	$\{(q_1, \epsilon), (q_1, \epsilon B)\}$
$\delta(q_1, *, B)$	$\{(q_1, \epsilon), (q_1, \epsilon B)\}$
$\delta(q_1, -, B)$	$\{(q_1, \epsilon), (q_1, \epsilon B)\}$
$\delta(q_1, /, B)$	$\{(q_1, \epsilon), (q_1, \epsilon B)\}$
$\delta(q_1, \epsilon, z_0)$	(q_f, ϵ)

State transition dgm



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Q.4 Multiplication of 2 no's

A) $m=2, n=2$

$B^m; n; BB...$

$B^{00}; 00; B...$

$\uparrow$
 $q_0 \rightarrow$

$B^*0; 00; B...$

$\uparrow$
 $q_1 \rightarrow$

$B^*0; 00; B...$

$\uparrow$
 $q_2 \rightarrow$

$B^*0; a0; B$

$\uparrow$
 $q_3 \rightarrow$

$B^*0; a0; B...$

$\uparrow$
 q_4

$B^*0; a0; 0B...$

$\uparrow$
 $\leftarrow q_5$

$B^*0; a0; 0B...$

$\uparrow$
 $q_2 \rightarrow$

$B^*0; aa; 0B...$

$\uparrow$
 $q_6 \rightarrow$

$B^*0; aa; 00B...$

$\leftarrow \uparrow q_5$

$B^*0; aa; 00B...$

$\uparrow$
 $q_2 \rightarrow$

$B^*0; aa; 00B...$

$\uparrow$
 $\leftarrow q_6$

$B^*0; 00; 00B$

$\uparrow$
 $q_0 \rightarrow$

$\vdots$

$B^{**}; 00; 0000B...$

$\uparrow$
 $q_0 \rightarrow$

$B^{**}; 00; 0000B...$

$\uparrow$
 $q_7 \rightarrow$

$B^{**}; 00; 0000B...$

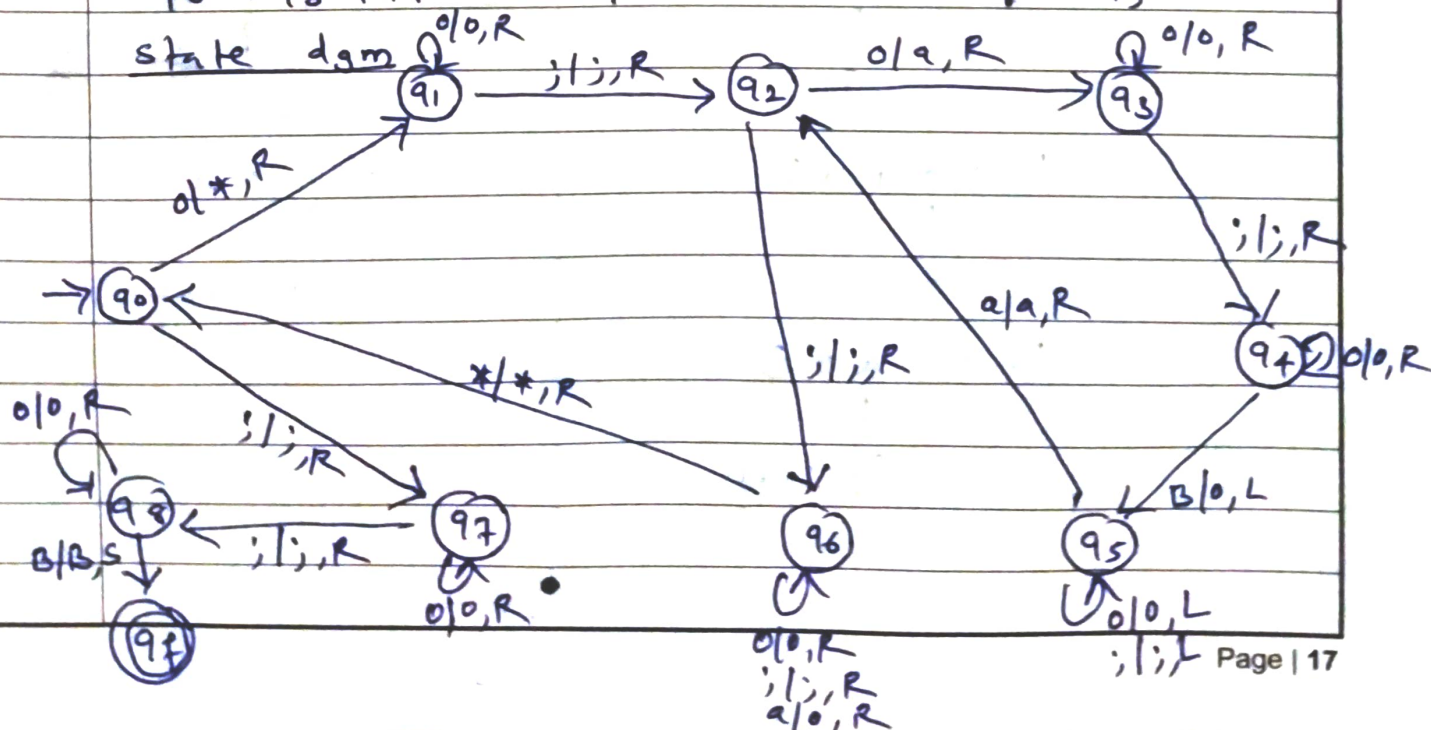
$\uparrow$
 $q_8 \rightarrow$

$B^{**}; 00; 0000B...$

$\uparrow$
 $q_f,$
stop

8:

Q	0	*	;	a	B
q0	q1 * R	-	q7 ; R	-	-
q1	q1 0 R	-	q2 ; R	-	-
q2	q3 a R	-	q6 ; L	-	-
q3	q3 0 R	-	q4 ; R	-	-
q4	q4 0 R	-	-	-	q5 0 L
q5	q5 0 L	-	q5 ; L	q2 a R	-
q6	q6 0 L	q0 * R	q6 ; L	q6 0 L	-
q7	q7 0 R	-	q8 ; R	-	-
q8	q8 0 R	-	-	-	qf BS



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Q.4

B)

TM for palindrome

Case 1: ~~odd~~^{Even} palindrome

B a b b a B

↑

 $q_0 \rightarrow$

B * b b a B

↑

 $q_1 \rightarrow$

B * b b a B

↑

B * b b a B

↑

 $q_0 \rightarrow$

B * * b * B

↑

 $q_1 \rightarrow$

B * * b * B

↑

 $\leftarrow q_5$

B * * * * B

↑

 $\leftarrow q_6$

B * * * * B

↑

 $q_0 \rightarrow$

B * * * * B

↑

 q_f ; stopCase 2: ~~Even~~ Odd palindrome

B a b a b a B

↑

 $q_0 \rightarrow$

B * * a * * B

↑

 $\leftarrow q_6$

B * * a * * B

↑

 $q_0 \rightarrow$

B * * * * * B

↑

 $q_1 \rightarrow$

B * * * * * B

↑

 $\leftarrow q_2$

B * * * * * B

↑

 q_f , stop

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Let $T = (Q, \Sigma, \Gamma, B, \delta, q_0, F)$ be the TM
 $Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_f \}$
 $\Sigma = \{ a, b \}$
 $\Gamma = \{ a, b, *, B \}$
 q_0 : initial state
 $F = \{ q_f \}$
 δ :

Q \ Γ	a	b	*	B
q_0	$q_1 * R$	$q_4 * R$	$q_f * S$	$q_f B S$
q_1	$q_1 a R$	$q_1 b R$	$q_2 * L$	$q_2 B L$
q_2	$q_3 * L$	$q_3 b S$	$q_f * S$	$q_f B S$ -
q_3	$q_3 a L$	$q_3 b L$	$q_0 * R$	-
q_4	$q_4 a R$	$q_4 b R$	$q_5 * L$	$q_5 B L$
q_5	$q_3 a S$	$q_6 * L$	$q_f * S$	-
q_6	$q_6 a L$	$q_6 b L$	$q_0 * R$	-

